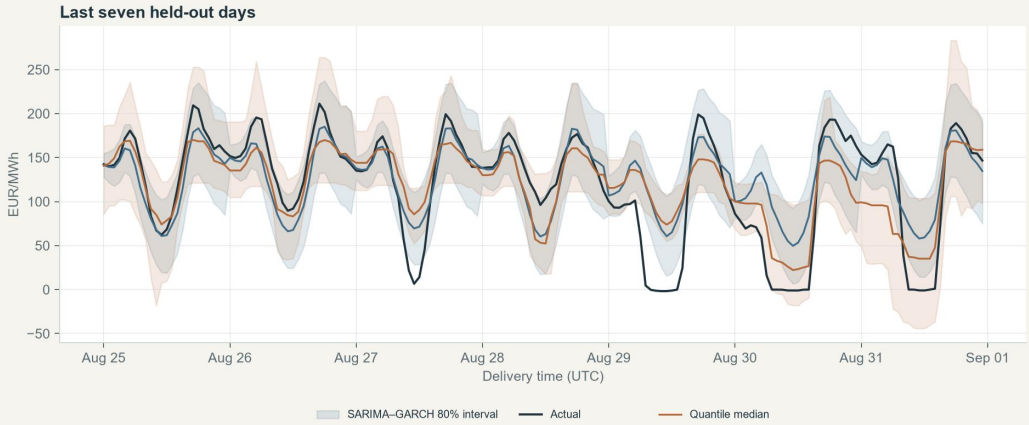
Day-ahead electricity price forecasting

Volatility and prediction intervals using SARIMA–GARCH and quantile gradient boosting



Germany–Luxembourg (DE-LU) · Hourly prices in UTC

520-day chronological holdout | 12,480 hourly forecasts

PRESENTED BY [STUDENT NAME] · USN [USN]

B. M. S. College of Engineering | Department of Machine Learning

A day-ahead forecast must describe both price and uncertainty

The question

Before delivery begins, what might each hour cost tomorrow?

01 **Estimate**

Produce 24 hourly point forecasts for one selected test day.

02 **Quantify**

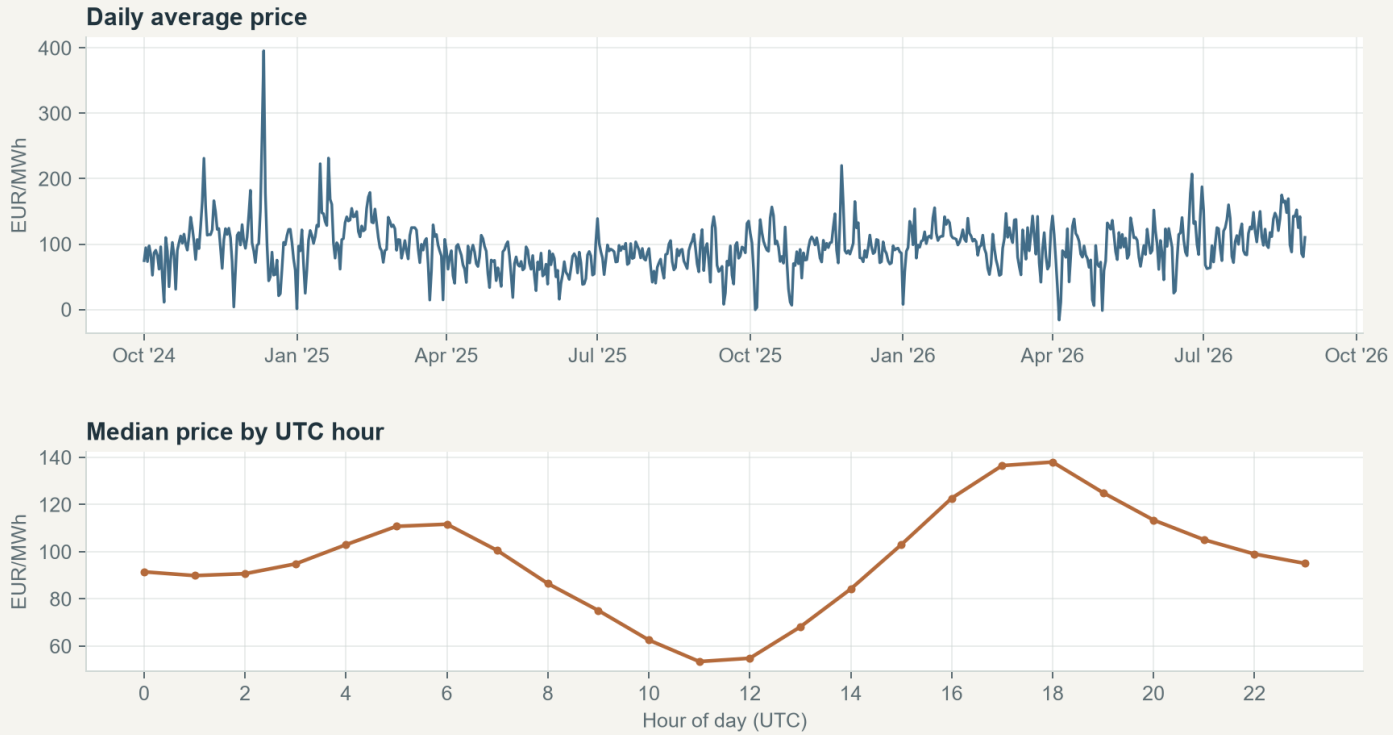
Show an 80% prediction interval, not just a single price.

03 **Evaluate**

Compare forecasts with held-out prices after the day is complete.

Application context: procurement, flexible demand and storage planning. The dashboard is an evaluation tool, not a trading system.

The study uses hourly DE-LU prices from a public market source



OBSERVATIONS

16,800

Hourly prices; timestamps stored in UTC.

COVERAGE

Oct 2024 – Aug 2026

The full series spans the training and test periods.

15-minute API observations are averaged to the hour. Negative prices are retained. Data: Energy-Charts price API; attribution to Bundesnetzagentur / SMARD, CC BY 4.0.

Prior work motivates the model choices and the comparison

PRICE DEPENDENCE

Autoregressive models capture useful intraday price structure. [1, 2]

CHANGING VARIANCE

Volatility and probabilistic forecasts matter alongside point error. [1, 3, 4]

REPRODUCIBLE EVALUATION

Benchmarking requires common test dates and consistent measures. [5]

Research question

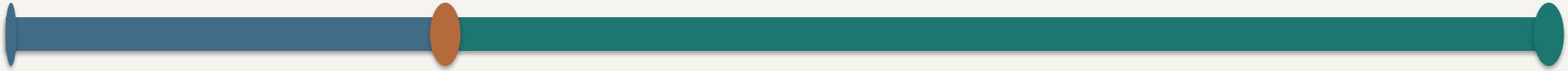
On the same long chronological DE-LU holdout, how do a seasonal SARIMA–GARCH interval and a direct quantile interval compare?

This is a focused classroom comparison, not a claim that the model combination is new or universally superior.

EVALUATION DESIGN

Both models are tested on the same future-only dates

Chronological split · no random shuffle



INITIAL TRAINING

180 days · 4,320 hourly observations

01 Oct 2024 – 29 Mar 2025

ROLLING HOLDOUT

520 days · 12,480 hourly forecasts

30 Mar 2025 – 31 Aug 2026 · 24 hours per forecast origin

01 **Forecast first**

Use information available before each delivery day.

02 **Update second**

Add that day's actuals only after forecasting.

03 **Score together**

Evaluate both models on the same 520 days.

MODEL 1 · BASELINE

SARIMA models the daily pattern; GARCH tracks residual volatility

SEASONAL MEAN

SARIMA(1,0,1) × (1,0,1,24)

An autoregressive and moving-average baseline with a 24-hour seasonal cycle, fit on the most recent 180 training days.

CONDITIONAL VARIANCE

$$\sigma^2_t = \omega + \alpha\varepsilon^2_{t-1} + \beta\sigma^2_{t-1}$$

GARCH(1,1) updates uncertainty from recent residuals. The nominal 80% interval uses forecast mean ± 1.645 standard deviations.

Assumption to check

Standardized innovations are treated as approximately normal for interval construction. Held-out coverage tests how well this assumption works.

Quantile boosting estimates the centre and both sides directly

τ = 0.10	τ = 0.50	τ = 0.90
LOWER QUANTILE	MEDIAN FORECAST	UPPER QUANTILE

10th–90th percentiles form the nominal 80% prediction interval.

INPUT FEATURES

Past prices at 24, 48 and 168 hours; recent 24- and 168-hour summaries; hour, weekday and month.

WHY QUANTILES?

Quantile loss allows an asymmetric interval without assuming normal forecast errors. Models use 150 shallow boosting trees.

Accuracy and interval quality answer different questions

POINT ERROR

MAE

Typical absolute miss; reported in EUR/MWh.

LARGE ERRORS

RMSE

Penalizes larger misses more strongly.

CALIBRATION

Coverage

Share inside a nominal 80% interval.

SHARPNESS

Width

Average interval size; smaller is not useful alone.

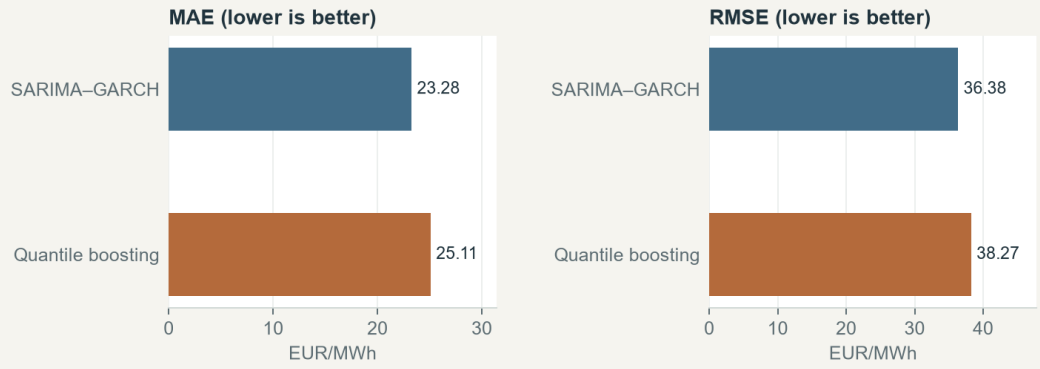
Additional checks

sMAPE is reported instead of MAPE because prices may be zero or negative. Pinball loss scores the lower and upper quantile forecasts.

Coverage is interpreted against 80%; it does not measure interval width or point accuracy.

On this holdout, SARIMA–GARCH has lower point error

MODEL	MAE	RMSE	sMAPE	80% COVERAGE	MEAN WIDTH	PINBALL 10	PINBALL 90
SARIMA–GARCH	23.28	36.38	83.3%	82.9%	83.25	6.43	6.30
Quantile boosting	25.11	38.27	86.5%	79.8%	81.90	6.77	6.09

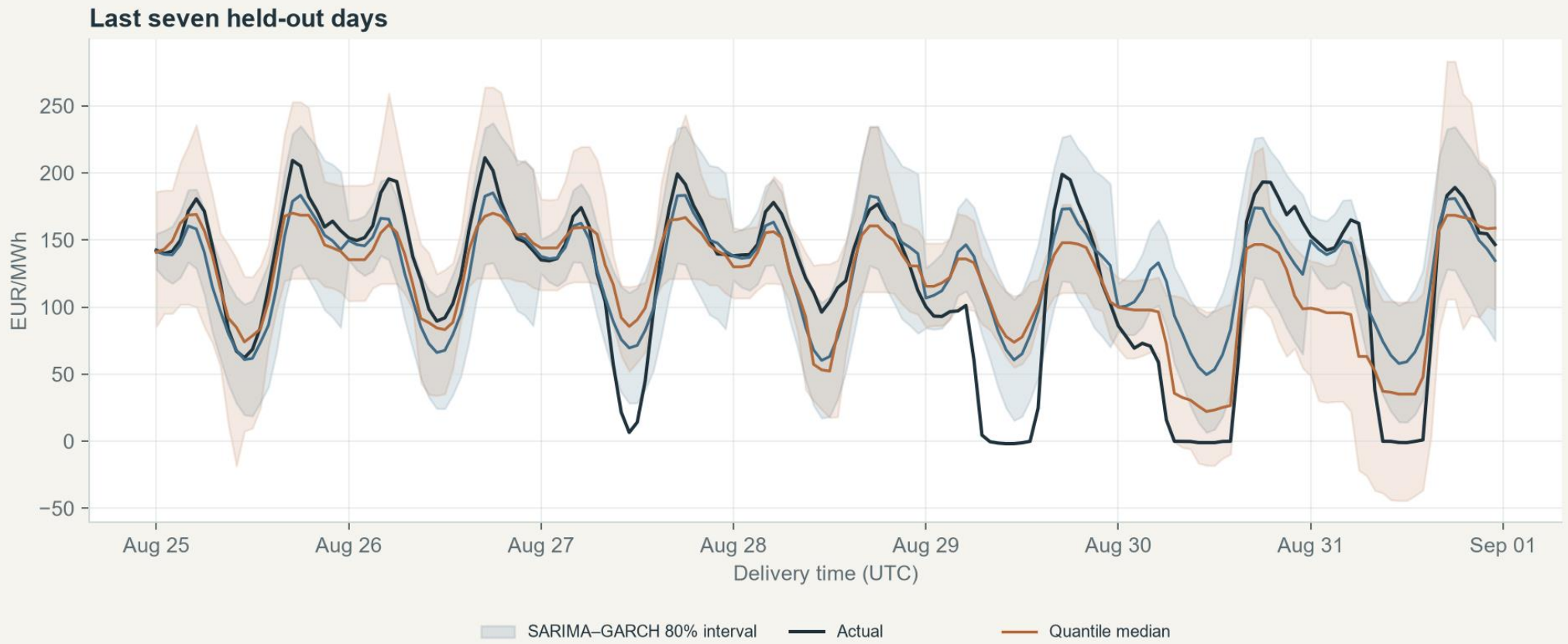


SARIMA–GARCH: lower MAE and RMSE.

Quantile boosting: coverage nearer 80% and slightly narrower intervals.

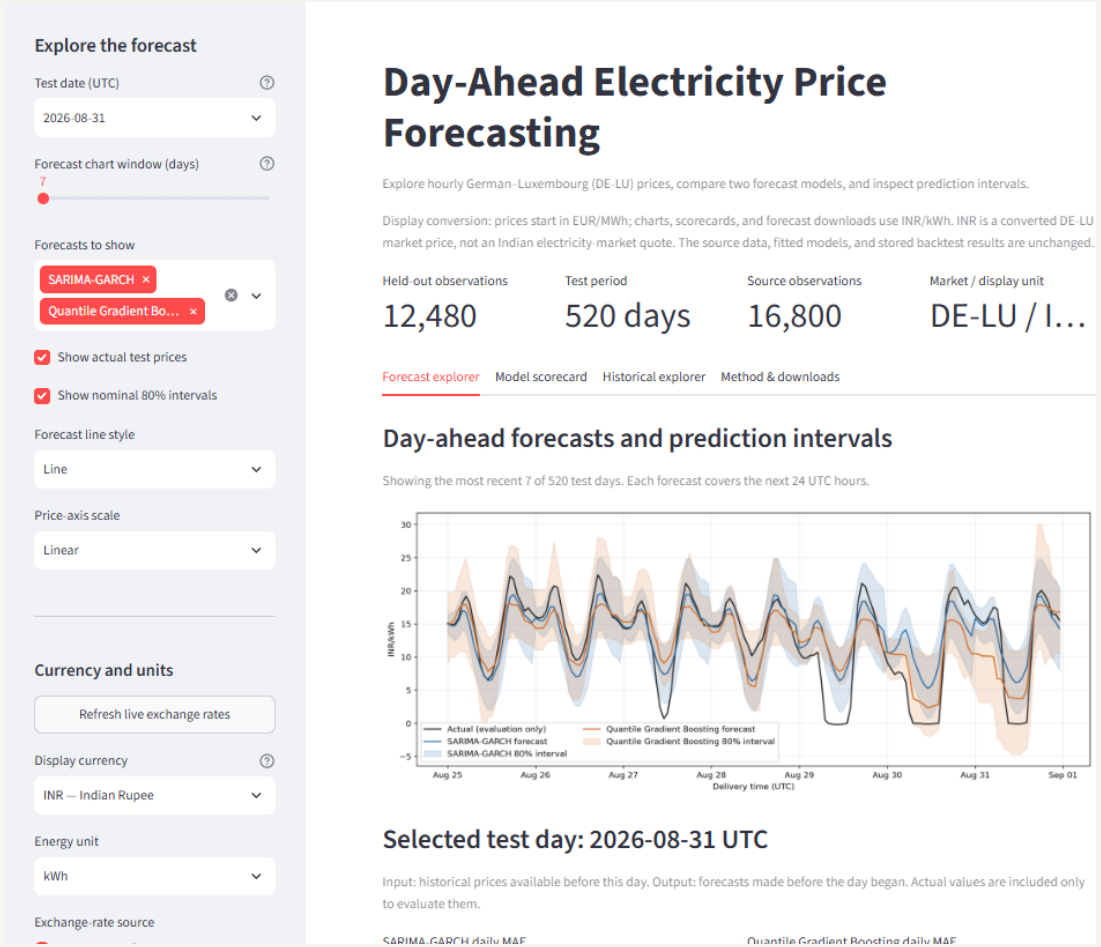
The preferred model depends on whether point error or interval behaviour matters more.

The final test week shows where point forecasts and intervals differ



Actual prices are shown only for retrospective evaluation. Shaded regions are nominal 80% intervals; sharp price moves remain difficult for both models.

The forecast explorer turns a test date into 24 hourly outputs



01 **Choose a test date**

The selector covers the 520 stored holdout dates, not arbitrary future dates.

02 **Inspect next-day estimates**

View 24 hourly forecasts, optional 80% bounds and the selected-day table.

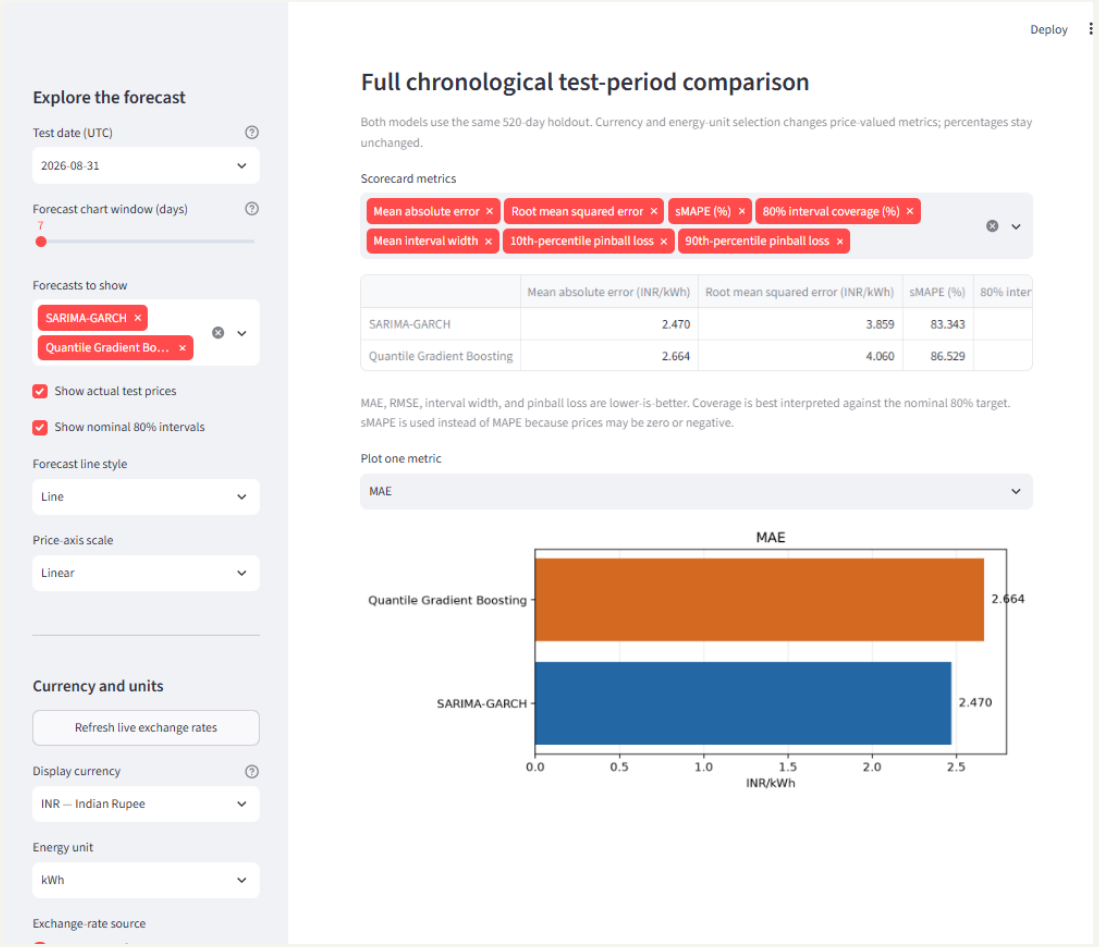
03 **Compare with actuals**

Held-out prices appear for evaluation only; they are not forecast inputs.

You can also show either model, hide actuals or intervals, choose line/step style, and use a linear or symmetric-log price axis.

LIVE DEMO: python -m streamlit run app.py

Currency and chart controls change the display, not the forecast



CURRENCY AND UNITS

Choose listed currencies or enter a three-letter code. INR/kWh is the default when live rates are available; EUR/MWh remains the source unit.

Rates are published ECB reference rates via Frankfurter. A positive manual rate is available; if rates are offline, EUR remains usable.

HISTORY AND DOWNLOADS

Explore hourly, daily or weekly prices, rolling averages, typical UTC-hour prices and price distributions. Download the visible history or selected-day forecasts as CSV.

INR is a display conversion of DE-LU prices—not an Indian electricity-market quote. Model inputs and stored results remain unchanged.

A selected day can be checked row by row

Timestamp (UTC)	Actual (INR/kWh)	SARIMA-GARCH forecast	SARIMA-GARCH lower 80%	SARIMA-GARCH upper 80%	Quantile Gradient Boosting forecast	Quantile Gradient Boosting low
2026-08-31 00:00:00+00:00	16.256	15.864	13.852	17.876	10.518	
2026-08-31 01:00:00+00:00	15.679	15.186	12.817	17.555	10.395	
2026-08-31 02:00:00+00:00	15.107	14.748	12.069	17.427	10.157	
2026-08-31 03:00:00+00:00	15.298	15.049	12.092	18.006	10.157	
2026-08-31 04:00:00+00:00	16.47	15.833	12.623	19.044	10.157	
2026-08-31 05:00:00+00:00	17.5	15.659	12.213	19.105	10.023	
2026-08-31 06:00:00+00:00	17.225	13.345	9.679	17.011	6.702	
2026-08-31 07:00:00+00:00	13.423	10.655	6.781	14.528	6.702	
2026-08-31 08:00:00+00:00	4.068	9.3	5.23	13.371	5.569	
2026-08-31 09:00:00+00:00	0.002	7.888	3.629	12.146	3.949	
2026-08-31 10:00:00+00:00	-0.006	6.799	2.36	11.237	3.877	
2026-08-31 11:00:00+00:00	-0.1	6.147	1.536	10.758	3.726	
2026-08-31 12:00:00+00:00	-0.109	6.286	1.508	11.064	3.726	
2026-08-31 13:00:00+00:00	-0.009	7.047	2.108	11.987	3.726	
2026-08-31 14:00:00+00:00	0.106	8.428	3.332	13.523	5.059	
2026-08-31 15:00:00+00:00	7.746	12.611	7.364	17.857	10.457	
2026-08-31 16:00:00+00:00	16.754	17.096	11.702	22.489	16.648	
2026-08-31 17:00:00+00:00	19.425	19.136	13.599	24.673	17.864	
2026-08-31 18:00:00+00:00	20.066	19.192	13.515	24.868	17.864	
2026-08-31 19:00:00+00:00	19.338	18.142	12.329	23.955	17.706	
2026-08-31 20:00:00+00:00	18.199	17.155	11.209	23.101	17.573	
2026-08-31 21:00:00+00:00	16.473	15.888	9.812	21.964	16.989	
2026-08-31 22:00:00+00:00	16.399	15.202	8.998	21.405	16.806	
2026-08-31 23:00:00+00:00	15.553	14.282	7.953	20.61	16.857	

24 HOURLY ROWS

Each row reports one UTC delivery hour. The selected display currency and unit appear in price-valued columns.

EVALUATION ONLY

Actual prices are included to measure the forecast after delivery. They are never used to build that same day's input features.

The daily CSV is downloadable in the selected display unit.

The result is useful within a clearly defined scope

WHAT THE TEST SHOWS

01 Point accuracy

SARIMA–GARCH has lower MAE and RMSE in this holdout.

02 Interval behaviour

Quantile boosting is closer to 80% coverage and slightly narrower.

LIMITS TO KEEP IN VIEW

One bidding zone and one historical period; no weather, demand or generation inputs; UTC days do not resolve local daylight-saving delivery days.

The dashboard's currency controls are for readability. An INR value is converted DE-LU data and should not be interpreted as an Indian market forecast.

Next: add external drivers, test more zones and seasons, calibrate intervals, and evaluate decision value.

References and data attribution

[1] Garcia, R. C. et al. (2005). A GARCH forecasting model to predict day-ahead electricity prices. IEEE Transactions on Power Systems. doi:10.1109/TPWRS.2005.846044

[2] Ziel, F. (2016). Forecasting electricity spot prices using Lasso. IEEE Transactions on Power Systems. doi:10.1109/TPWRS.2016.2521545

[3] Nowotarski, J. & Weron, R. (2018). Recent advances in electricity price forecasting: probabilistic forecasting. Renewable & Sustainable Energy Reviews. doi:10.1016/j.rser.2017.05.234

[4] Uniejewski, B., Weron, R. & Ziel, F. (2018). Variance stabilizing transformations for electricity spot price forecasting. IEEE Transactions on Power Systems. doi:10.1109/TPWRS.2017.2734563

[5] Lago, J. et al. (2021). Forecasting day-ahead electricity prices: algorithms, best practices and open-access benchmark. Applied Energy. doi:10.1016/j.apenergy.2021.116983

Data: Energy-Charts price API, DE-LU zone; attribution to Bundesnetzagentur / SMARD; CC BY 4.0. Exchange-rate service: Frankfurter, which distributes ECB reference rates.

All reported model results are computed from the original EUR/MWh backtest. Dashboard currency conversion affects presentation only.